

BTC volatility at the 5-minute horizon and its link to ES

Return anatomy, vol persistence, market-open effects and the ES-BTC coupling, evaluated for the Polymarket 5-minute BTC up/down market. Binance BTCUSD spot 1-second klines and E-mini S&P 500 (ES) 1-second BBO, 16 May 2023 – 12 September 2026.

Purpose. The Polymarket 5-minute BTC up/down market pays on the sign of the BTC/USD move over a clock-aligned five-minute window. Its fair price is a statement about the distribution of the *next* five-minute return, so every question here is asked at short horizons and in the same clock-aligned windows the market uses. The ES futures contract is examined as a possible external signal.

Headline findings.

1. **The sign of the 5-minute return is essentially unpredictable from past returns.** $P(\text{up}) = 0.498$; autocorrelation of the return at lag 1 is -0.016 ; $P(\text{up} \mid \text{last window up}) = 0.492$ versus 0.505 after a down window. The reversal is statistically real but worth about one cent per dollar of contract.
2. **Many windows are decided by noise.** 10.7% of windows finish within 1 bp of where they started, 20.4% within 2 bp and 44.5% within 5 bp. The typical (median) window is far calmer than the standard deviation suggests: excess kurtosis is 104.
3. **Vol level, by contrast, is highly predictable.** The next window's log realised variance is forecast with out-of-sample $R^2 = 0.59$ from BTC's own recent history (HAR); the last 30 minutes alone give 0.56. After a top-decile window the next window's vol averages $2.2\times$ its seasonal norm (median $1.6\times$); after a bottom-decile window, $0.37\times$ (median $0.26\times$).
4. **BTC vol has long memory with a fast component.** The autocorrelation of 1-minute log variance is 0.51 at one minute, still 0.36 at one hour and 0.23 at one day; a two-exponential fit has a 54-minute fast half-life (32% weight) and a 1.7-day slow one. Extreme vol shocks decay as a power law and leave vol several times its baseline for hours, whether or not ES is trading.
5. **The US session, not the clock, drives BTC's intraday shape.** Weekday variance is about $2.5\text{--}3\times$ weekend variance through the European morning, $25\times$ at 08:30 (US data), $7\times$ for the hour after the 09:30 cash open, and $\approx 1\times$ overnight. BTC's per-minute vol while ES trades is 43% annualised versus 29% when it does not.
6. **ES vol adds almost nothing to a BTC vol forecast.** Conditioning on ES's last-minute move or last-5-minute vol produces large raw differences (BTC vol $2.3\times$ normal after a >5 sd ES minute), but once BTC's own last five minutes are in the regression, ES raises R^2 from 0.620 to 0.623. Same-window vol surprises correlate at 0.17 (levels: 0.30); cross-correlation at one window's lead in either direction is ≈ 0.03 .
7. **Returns co-move strongly and contemporaneously.** β of BTC on ES is 0.80 at 10 seconds, 1.02 at 5 minutes and 1.16 at 1 hour, with correlation $0.31 \rightarrow 0.40 \rightarrow 0.46$. Lead-lag bars at 10 s and 1 min are small and symmetric: there is no measurable lead of ES over BTC at this resolution. When ES moves more than 3 sd in a window, BTC has the same sign 92% of the time — but within the same window.
8. **Negative result for a mid-window entry rule: ES's move so far in a window says nothing useful about BTC's remaining move.** β of BTC's remaining return on ES's first-minute return is 0.06 ($t = 5.0$, $R^2 = 0.0002$); $P(\text{remaining BTC move has ES's})$

sign) = 0.501, and 0.495 after a >2 sd ES move. BTC's own move so far is likewise uninformative ($\beta \approx -0.02$). What a mid-window entrant knows is the current price relative to the window open and the (well-forecastable) remaining vol; that is the whole edge.

1. Data and method

1.1 Inputs

Two raw sources are used. **BTC** is Binance BTCUSDT spot, 1-second klines, from 16 May 2023 (the first month on disk) to 12 September 2026. Binance spot stands in for the Chainlink BTC/USD reference on which the Polymarket market actually settles; the two track each other closely but are not the same series (see the caveats in section 9). **ES** is the E-mini S&P 500 front-month future from Databento's bbo-1s feed. The ES price used throughout is the size-weighted mid ("microprice"), $(\text{bid}_{\text{size}} \cdot \text{ask}_{\text{px}} + \text{ask}_{\text{size}} \cdot \text{bid}_{\text{px}}) / (\text{bid}_{\text{size}} + \text{ask}_{\text{size}})$, which moves inside the 0.25-point tick as queue sizes change and so is much less discretised than the plain mid. Three derived inputs are built from these once and cached (Table 1).

series	minutes	span	note
BTC (Binance BTCUSDT spot, 1 s klines)	1,750,394	2023-05-16 → 2026-09-12	24/7; ES is quoted in 67.0% of these minutes
ES (front-month, Databento bbo-1s)	1,173,390	2023-05-16 → 2026-09-11	Globex hours; 17:00–18:00 ET break removed
10-second grid (BTC close + ES microprice)	10,502,362 rows	2023-05-16 → 2026-09-12	ES present in 56.8% of grid points

Table 1 – The three derived inputs. Annualised BTC vol from the mean per-minute variance: 38.8% over all minutes, 43.0% while ES is quoted, 28.6% while it is not.

Times are US Eastern (ET, with daylight-saving shifts), because the events that matter — US data at 08:30, the cash open at 09:30, the 10:00 releases, the 16:00 close, the CME maintenance break at 17:00 and the Globex reopen at 18:00 — are fixed in that clock. Minutes falling in the 17:00–18:00 break are removed from the ES series (the sparse pre-open indications quoted there are not tradeable prices). "Weekend" means Saturday or Sunday in ET. "ES open" for a BTC minute means ES was quoted in that minute.

1.2 Realised variance and deseasonalisation

The **realised variance** (RV) of a minute is the sum of squared 1-second log returns within it, computed from consecutive 1-second closes (BTC) or microprices (ES) with no gap longer than two minutes and no contract change. RV over a 5-minute window is the sum of its five minute-RVs. Minutes with fewer than 20 one-second returns, and windows with fewer than 200, are dropped. Vol is quoted annualised: for BTC on a 24/7 clock, $\sqrt{(\text{RV} \times 365.25 \times 1440)}$ for a per-minute RV or $\sqrt{(\text{RV} \times 365.25 \times 288)}$ for a 5-minute RV, so that 0.40 means 40% a year; ES, where compared directly to BTC in section 4, uses a trading-time convention of 252 days $\times$ 1,380 minutes, and where it must be on the same footing as BTC (section 7) the 24/7 clock is used and labelled.

Both series have a strong intraday shape, so most of the analysis works on the **deseasonalised variance ratio** $x = \text{RV} / (\text{mean RV at that minute of day})$. For BTC the mean is taken separately for weekdays and weekends, so that the weekend's different profile does not leak into weekday ratios. By construction x averages 1 ("baseline"); a window with $x = 4$ has twice the vol that is normal for that time of day. Because RV is heavily right-skewed, $\log x$

is used for autocorrelations and regressions, and the *median* of x sits well below 1 (a typical window is calmer than the average window).

1.3 Autocorrelation fits and event studies

The autocorrelation function (ACF) $\rho(k)$ of $\log x$ is fitted on 300 log-spaced lags between 1 and three days with three shapes: a single exponential $a \cdot e^{-k/\tau}$ (one half-life $\tau \cdot \ln 2$), a power law $c \cdot k^{-\alpha}$ (long memory: no single time-scale, ρ halves each time the lag grows by a factor $2^{1/\alpha}$), and the sum of two exponentials (a fast and a slow component). R^2 is on the log-spaced lags, so it weights the short and long ends comparably. An **event study** selects the minutes in which x exceeds its 99th (or 99.9th) percentile and averages x — of the same or the other series — at each offset from 30 minutes before to 120 or 720 minutes after. The mean minus 1 is the “excess”. Because the selection picks the peak of a cluster, the minutes just before a “shock” are also elevated; that is a feature of the selection, not evidence of anticipation.

1.4 Forecasting: HAR, AR(1) and innovations

The **HAR** (heterogeneous autoregressive) regression forecasts the mean of $\log x$ over the next horizon from trailing means of $\log x$ over the last window, the last 30 minutes, 2 hours, 24 hours and one week. It is the standard parsimonious way to capture long-memory vol with a handful of coefficients. It is compared with AR(1) (last window only) and with the last-30-minute mean alone. Coefficients are estimated on the first 70% of the sample; out-of-sample R^2 is $1 - \text{SSE}/\text{SST}$ on the last 30%, with SST taken around the training-sample mean, so a value near the in-sample one means the fit generalises.

Section 6 needs the part of each series' vol that was *not* predictable from its own past — its **innovation**. Two versions are used: the residual of an AR(12) regression of $\log x$ on its own last twelve windows (one hour), and the residual of a HAR regression with last-window, 30-minute, 2-hour and 24-hour means. A third, cruder version is the first difference $\Delta \log x$. Correlating innovations answers “when ES vol surprises, does BTC vol surprise in the same window?”, which a correlation of levels cannot, because levels of two persistent series share slow regimes.

1.5 Beta and cross-correlation

β of BTC on ES over a horizon h is $\text{cov}(r_{\text{BTC}}, r_{\text{ES}})/\text{var}(r_{\text{ES}})$ using log returns over consecutive h -intervals of the 10-second grid; $\beta = 1$ means a 1 bp ES move is matched on average by a 1 bp BTC move. Correlation and $R^2 = \text{corr}^2$ say how much of BTC's variance that relationship accounts for. The **lead-lag** version regresses $\text{BTC}(t)$ on $\text{ES}(t-k)$: a large bar at $k > 0$ means an ES move is followed by a BTC move k intervals later (ES leads); at $k < 0$, BTC leads. The **cross-correlation function** (CCF) of section 6 is the same idea applied to standardised vol innovations, with positive lag meaning ES leads. With $n \approx 219,000$ windows, two standard errors for a correlation are about 0.004, so almost everything is “significant”; the question throughout is whether an effect is large enough to matter.

2. The 5-minute BTC return

Returns are computed between the first and last 10-second grid points of each clock-aligned window (:00, :05, ...), which is how the market is defined. Table 2 collects the unconditional facts; Figure 1 shows the vol profile through the day.

quantity	value
number of clock-aligned 5-minute windows	350,078
standard deviation of r_5	0.00145 (14.5 bp) – annualised 47.1%
skew / excess kurtosis	–1.03 / 104.5
$P(r_5 > 0)$	0.4982
$P(r_5 = 0 \text{ exactly})$	0.36%
$P(r_5 < 1 \text{ bp})$	10.7%
$P(r_5 < 2 \text{ bp})$	20.4%
$P(r_5 < 5 \text{ bp})$	44.5%
$P(r_5 < 10 \text{ bp})$	69.7%
autocorrelation of r_5 , lags 1...4	–0.016, –0.014, –0.000, –0.005
autocorrelation of $\text{sign}(r_5)$, lags 1...4	–0.013, –0.019, –0.012, –0.011
$P(\text{up} \mid \text{previous window up}) / P(\text{up} \mid \text{previous window down})$	0.4918 / 0.5046 (2 s.e. ≈ 0.0034)

Table 2 – Anatomy of the 5-minute BTC log return r_5 (window open to window close on the 10-second grid), all windows, weekdays and weekends. Lags 5–12 of both autocorrelation functions are all within ± 0.009 .

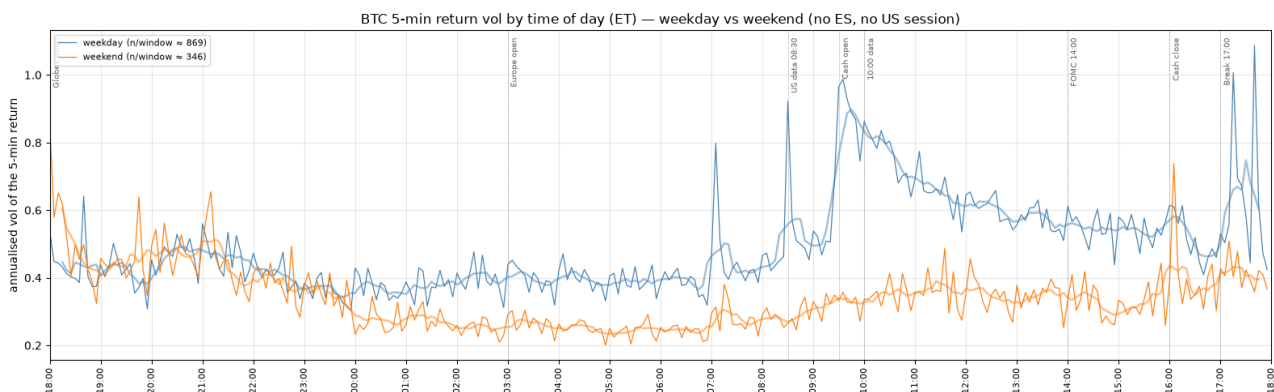


Figure 1 – Annualised vol of the 5-minute return by time of day (ET), weekdays (blue) versus weekends (orange), shown from the 18:00 Globex open round to the next 17:00. Thin lines are the raw per-window standard deviations (≈ 869 weekday and ≈ 346 weekend observations per window); thick lines are a six-window moving average. Dotted verticals mark the usual US events.

Interpretation. The 5-minute return has a standard deviation of 14.5 bp, or 47% annualised — noticeably above the 39% implied by the mean 1-second realised variance, which together with the excess kurtosis of 104 says that the standard deviation is dominated by a small number of very large windows. The centre of the distribution is tight: 45% of windows move less than 5 bp end to end, one in five less than 2 bp and one in ten less than 1 bp. A 1 bp move on a \$60,000 price is \$6; at that scale the sign of the resolution is determined by which trades happened to print at the window boundaries, and by any difference between the Binance price used here and the Chainlink reference the market settles on. That fraction of windows is effectively a coin flip whatever the model says. Exact ties (0.36% of windows) are rare but not negligible if the market's tie rule is unfavourable.

Directional information in past returns is close to nil. The lag-1 autocorrelation of the return is -0.016 and of its sign -0.013 (2 s.e. ≈ 0.003), and the conditional up-probabilities are 0.492 after an up window and 0.505 after a down window. The effect is statistically detectable on 350,000 windows but economically tiny: acting on it moves the fair probability by about 1.3 points, less than a typical bid-ask spread in a binary market. Nothing beyond lag 2 is

distinguishable from zero. The negative skew (-1.03) says the largest windows are more often down than up, but this does not translate into a usable bias in $P(\text{up})$, which is 0.498 .

The vol profile is the first strong structure. On weekdays, vol sits near 35–40% annualised through the Asian and European sessions, climbs from about 07:00, spikes at 08:30 (US data releases; $\approx 90\%$ annualised in that window), peaks in the 09:30–10:00 cash-open window at just over 100%, and decays through the afternoon to $\approx 55\%$ before spikes at the 16:00 close and around 17:00–18:00. The weekend line has none of this: it drifts between 25% and 45% with no structure at US times, apart from isolated spikes (near 16:00, and around 20:00–21:00) that, with only ≈ 346 observations per window, are most likely single-event days rather than recurring features. In per-window terms a quiet weekday window has a standard deviation of ≈ 11 bp, the cash-open window ≈ 31 bp and a weekend window ≈ 8 bp — a factor of four in the scale of the move the market is pricing, before any conditioning on recent vol.

3. BTC vol autocorrelation at short horizons

This section asks how far ahead BTC vol is knowable from BTC vol alone. Figure 2 shows the ACF of deseasonalised log variance at 1-minute and 5-minute resolution out to three days, with the three fitted shapes; Table 3 gives the fits.

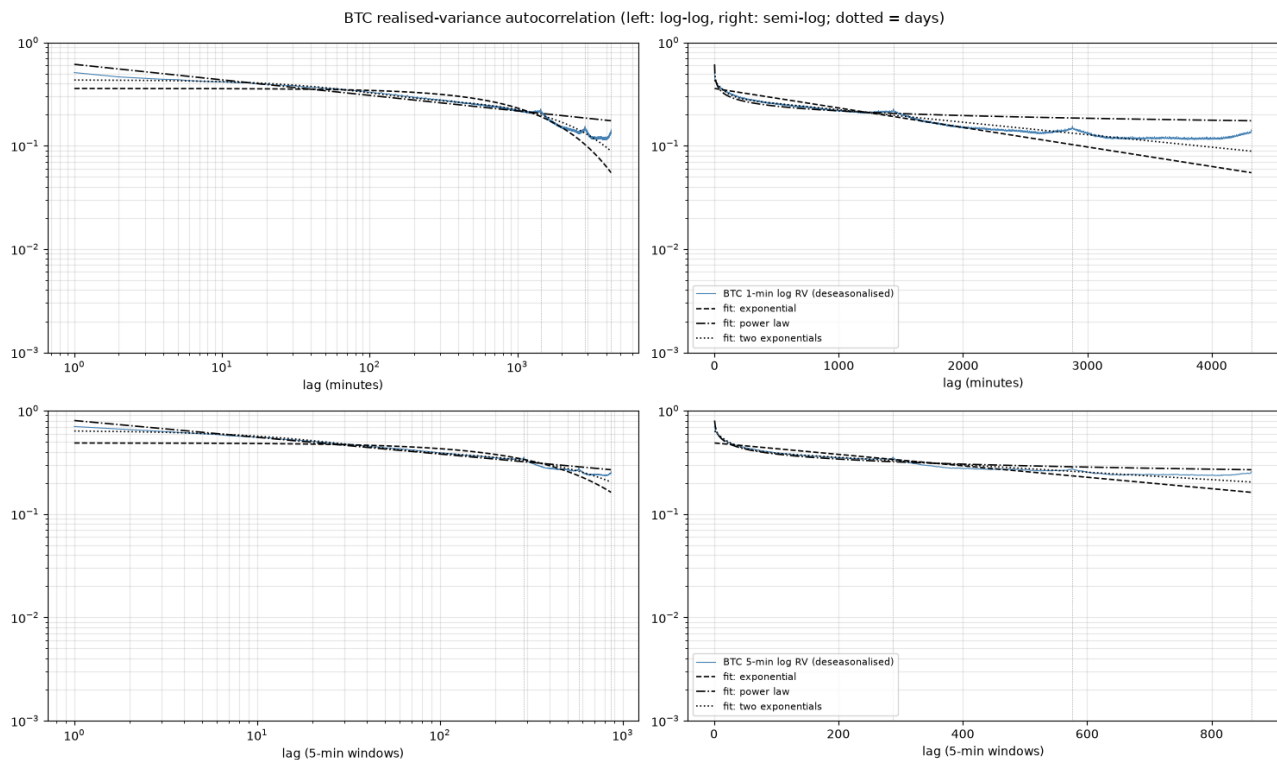


Figure 2 – Autocorrelation of deseasonalised log realised variance, 1-minute (top) and 5-minute (bottom) resolution, on log-log (left) and semi-log (right) axes; dotted verticals at multiples of one day. Dashed: exponential; dash-dot: power law; dotted: two exponentials. A straight line on the left panels is a power law; on the right, an exponential.

resolution	model	R ² (log-spaced lags)	fitted parameters	implied half-life
1-minute	exponential	0.861	0.3576, 2302	1595.7 min
1-minute	power law	0.901	0.6121, 0.1502	$\alpha = 0.150$: ρ halves every x101.1 in lag
1-minute	two exponentials	0.986	0.1398, 78.34, 0.2936, 3603	fast 54.3 min (weight 32%), slow 2497 min
5-minute	exponential	0.810	0.4867, 785.1	544.2 windows
5-minute	power law	0.957	0.8018, 0.1621	$\alpha = 0.162$: ρ halves every x71.9 in lag
5-minute	two exponentials	0.984	0.2274, 25.25, 0.4166, 1208	fast 17.5 windows (weight 35%), slow 837 windows

Table 3 – Fits to the autocorrelation of deseasonalised log realised variance, lags 1 to 3 days (4,320 minutes or 864 windows). Parameters are a , τ (exponential $a \cdot e^{-k/\tau}$); c , α (power law $c \cdot k^{-\alpha}$); a_1 , τ_1 , a_2 , τ_2 (two exponentials). τ in minutes for the 1-minute series and in 5-minute windows for the 5-minute series.

Interpretation. BTC vol is strongly and persistently autocorrelated. At 1-minute resolution ρ is 0.51 at lag 1, 0.44 at 5 minutes, 0.38 at 30 minutes, 0.36 at one hour and still 0.23 at one day; at 5-minute resolution the corresponding figures are 0.70 at one window, 0.59 at 30 minutes, 0.54 at one hour and 0.36 at one day. The shape is not a single exponential (R^2 0.81–0.86, and the dashed line visibly falls too fast at long lags): it is closest to a power law over most of the range (R^2 0.90 and 0.96) and best described by two exponentials (R^2 0.986 and 0.984). The two-exponential decomposition is the most useful summary — a fast component with a half-life of about 54 minutes (17.5 windows $\approx$ 88 minutes at 5-minute resolution), carrying roughly a third of the weight, on top of a slow component with a half-life of 1.7–2.9 days. The residual bumps at exactly one and two days in the 1-minute ACF are leftover daily seasonality (the deseasonalisation removes the mean profile, not day-to-day variation in its amplitude). The practical reading: a shock to BTC vol is still mostly present an hour later and about a third of it persists into the next day.

3.1 How predictable is the next window's variance?

horizon	model	R ² in-sample	R ² out-of-sample	β last 5 min	β last 30 min	β last 2 h	β last 24 h	β last week
next 5min	HAR	0.593	0.590	0.270	0.401	0.186	0.102	0.030
next 5min	AR(1)	0.496	0.487	0.705	–	–	–	–
next 5min	last 30min only	0.560	0.555	–	0.884	–	–	–
next 15min	HAR	0.699	0.705	0.209	0.379	0.227	0.130	0.039
next 15min	AR(1)	0.559	0.549	0.666	–	–	–	–
next 15min	last 30min only	0.662	0.664	–	0.856	–	–	–
next 1h	HAR	0.739	0.767	0.128	0.300	0.279	0.203	0.065
next 1h	AR(1)	0.539	0.544	0.594	–	–	–	–
next 1h	last 30min only	0.682	0.710	–	0.789	–	–	–
next 24h	HAR	0.674	0.677	0.029	0.065	0.226	0.266	0.315
next 24h	AR(1)	0.357	0.376	0.387	–	–	–	–
next 24h	last 30min only	0.491	0.522	–	0.536	–	–	–

Table 4 – Forecasting the mean deseasonalised log realised variance over the next horizon from the last window (AR(1)), the last 30 minutes only, or the HAR set of five trailing means. Coefficients are fitted on the first 70% of the sample; out-of-sample R^2 is measured on the final 30%.

Interpretation. Five trailing means of log variance explain 59% of the variation in the next window's log variance out of sample, with no sign of overfitting (in-sample 0.593 versus out-of-sample 0.590 on the last 30% of the sample, and the same closeness at every horizon). The last window alone gives 0.49; the last 30 minutes alone gives 0.56, so a single half-hour average captures most of the forecastable part. The HAR coefficients for the next-window target — 0.27 on the last window, 0.40 on the last 30 minutes, 0.19 on the last two hours, 0.10 on the last day, 0.03 on the last week — sum to almost one, the signature of near-unit-root persistence in log vol, with the weight concentrated on the last half-hour. R^2 rises with the horizon (0.71 for the next 15 minutes, 0.77 for the next hour) because the target averages over more windows and so contains less measurement noise; for the next day the weight shifts to the daily and weekly means, as one would expect. An R^2 of 0.59 in log variance still leaves a large unexplained residual in every window: vol is predictable in the sense of being strongly state-dependent, not in the sense of being known.

decile of current window	next window (mean)	next (median)	+2 windows (mean)	+2 (median)	+30 min (mean)	+30 min (median)	+1 h (mean)	+1 h (median)	+1 day (mean)	+1 day (median)
D1	0.37	0.26	0.41	0.28	0.46	0.29	0.51	0.31	0.72	0.39
D2	0.47	0.36	0.51	0.38	0.57	0.39	0.61	0.41	0.77	0.46
D3	0.54	0.44	0.58	0.44	0.64	0.46	0.68	0.47	0.80	0.51
D4	0.61	0.50	0.65	0.51	0.70	0.52	0.75	0.53	0.84	0.55
D5	0.68	0.57	0.72	0.58	0.77	0.58	0.81	0.59	0.90	0.60
D6	0.76	0.65	0.79	0.64	0.85	0.64	0.90	0.64	0.95	0.64
D7	0.86	0.74	0.89	0.73	0.94	0.72	0.98	0.71	1.03	0.69
D8	1.00	0.86	1.03	0.84	1.06	0.82	1.06	0.80	1.09	0.75
D9	1.21	1.07	1.23	1.03	1.23	0.98	1.25	0.94	1.23	0.84
D10	2.17	1.62	2.08	1.53	1.93	1.37	1.80	1.27	1.44	1.01

Table 5 – Vol of a later 5-minute window as a multiple of its seasonal baseline, given the decile of the current window's deseasonalised variance (D1 = calmest tenth, D10 = wildest). Entries are the square root of the mean or median variance ratio, so they read as vol multiples; 1.00 = the average window at that time of day.

Interpretation. Table 5 is the same information in the form a trader can use. Given only the decile of the current window's variance, the next window's vol relative to the seasonal norm ranges from $0.37\times$ (bottom decile) to $2.17\times$ (top decile) on a mean basis, or $0.26\times$ to $1.62\times$ on a median basis — a six-fold spread. The multiple decays towards 1 slowly: after a top-decile window the next hour is still at $1.8\times$ (median $1.3\times$) and the next day at $1.4\times$ (median $1.0\times$). Two features deserve note. First, the mean and median columns differ substantially because x is right-skewed; for pricing a binary on the sign of the move, the relevant scale is closer to the median of the variance distribution than to its mean, and even the fifth-decile row has a median multiple of only 0.57. Second, the “1.00 = baseline” line is crossed only at decile 8, which again reflects that the average window is a wild one; the typical window is at 0.6–0.7 of the mean-based baseline. A window-by-window vol estimate — seasonal norm $\times$ current-state multiple — is the first input to any fair price for the market, and this table says the multiple is both large and estimable.

3.2 Decay of extreme vol shocks

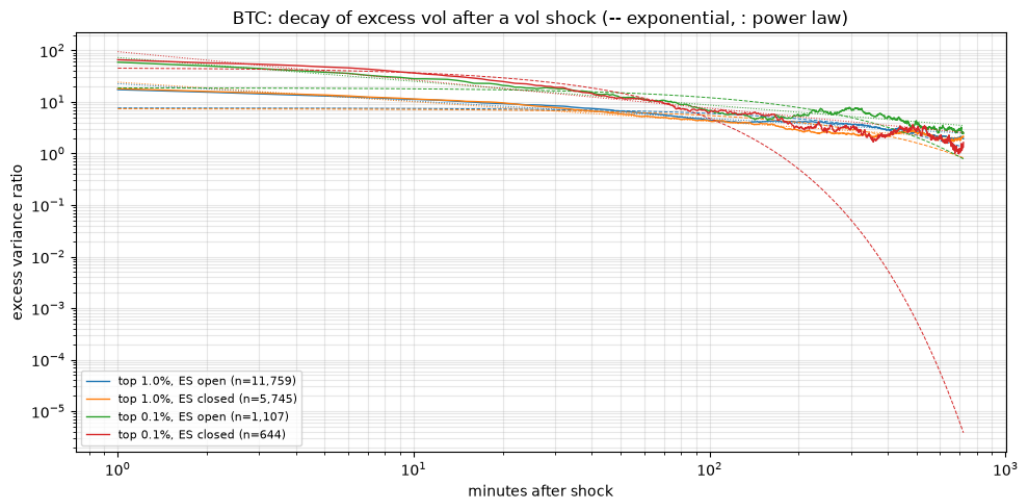


Figure 3 – Mean excess deseasonalised variance ($x - 1$) in the minutes after the top-1% and top-0.1% BTC variance minutes, split by whether ES was quoted at the time, log-log axes. Dashed lines are exponential fits, dotted are power-law fits.

shock set	n	x at k = 0	x at k = -1	k = +1	k = +5	k = +30	k = +120	exp. fit half-life (min)	power-law α
top 1.0%, ES open	11,800	24.7	18.9	18.4	14.1	9.39	5.15	285	0.336
top 1.0%, ES closed	5,740	27.1	19.8	19.1	14.7	8.76	4.95	227	0.38
top 0.1%, ES open	1,110	85.6	58.8	60	38.7	19.4	6.77	157	0.461
top 0.1%, ES closed	644	93.2	64.8	66.8	48.8	20.2	6.91	30.6	0.549

Table 6 – Mean deseasonalised BTC variance ratio x around the shock minute ($k = 0$), for the top 1% and top 0.1% of minutes, split by whether ES was quoted at the time. $x = 1$ is the time-of-day baseline. The notebook also reported the first minute at which the excess fell below 0.1; it was undefined for every set because the excess never fell that far within the 720-minute window.

Interpretation. A top-1% minute has variance $\approx 25\times$ its baseline; a top-0.1% minute $\approx 85\text{--}95\times$. The minute after is already down to $\approx 18\times$ ($\approx 60\text{--}67\times$ for the extreme set), i.e. the sharpest part of the decay happens within a minute; thereafter the excess falls as a power law ($\alpha \approx 0.34\text{--}0.55$) rather than exponentially — the exponential fits (dashed) collapse far too fast, and the fitted half-lives in Table 6 should not be taken literally (the 31-minute value for the smallest set is a fit to the first few points only). Half an hour after a top-1% shock vol is still $3\times$ baseline ($\sqrt{9.4}$), two hours later $2.3\times$ ($\sqrt{5.2}$), and the curves in Figure 3 have not returned to baseline after 12 hours. The **ES-open and ES-closed responses are almost identical** in level and shape: BTC's vol dynamics after its own shocks do not depend on whether ES is trading. Equally telling, the minute before the shock ($k = -1$) is as elevated as the minute after — a shock is typically the peak of a build-up, not a bolt from a calm sky. For the market this means: after a violent minute, expect vol to remain several times normal for the rest of the hour, and price the next windows accordingly.

4. BTC vol around market opens

The weekend, with neither ES nor US equities trading, is the natural control for whether BTC's intraday shape is driven by the clock or by the US market. Figure 4 shows per-minute

BTC vol for weekdays and weekends with ES alongside, and the weekday/weekend variance ratio minute by minute; Figure 5 and Table 7 zoom in on the two hours after each of five events, for BTC and ES side by side.

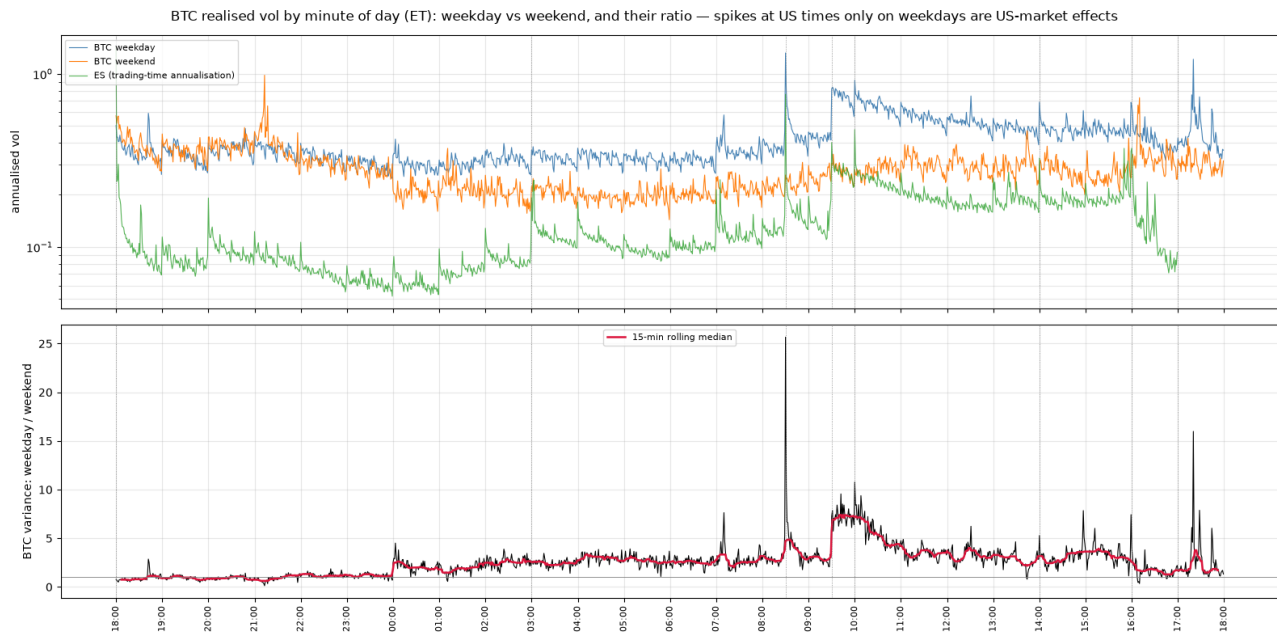


Figure 4 – Top: annualised vol from the mean per-minute realised variance by minute of day (ET), BTC weekdays (blue), BTC weekends (orange) and ES (green, trading-time annualisation), log scale. Bottom: the ratio of weekday to weekend mean BTC variance (black) and its 15-minute rolling median (red).

Interpretation. The ratio isolates the US-market effect cleanly. From 00:00 to 08:00 ET weekday variance is $2\text{--}3\times$ weekend variance — BTC is simply busier on trading days, with a mild rise through the European morning. Then the US events show up as ratios that no clock effect could produce: ≈ 25 at 08:30 (the data-release minute; BTC's weekday variance in that minute is $25\times$ its weekend variance at the same time of day), a plateau of ≈ 7 for the whole 09:30–10:30 hour after the cash open, a slow decline to $\approx 3\text{--}4$ through the afternoon, and renewed spikes at 16:00 and around 17:00–17:30. The Europe open at 03:00 is barely visible in either line. Between 18:00 and midnight the ratio is ≈ 1 , but this segment should be read with care: with weekends defined by ET calendar day, Friday evening (ES closed) counts as weekday and Sunday evening (ES open from 18:00) counts as weekend, so both groups are mixed in those hours and the step at 00:00 is at least partly a classification artefact rather than a genuine midnight effect. Isolated weekday spikes such as the one near 07:00 and the pair at 17:15–17:30 sit at no scheduled US release; a mean over ≈ 870 weekdays can be moved by a handful of extreme days, and these are best treated as unexplained until checked against specific dates.

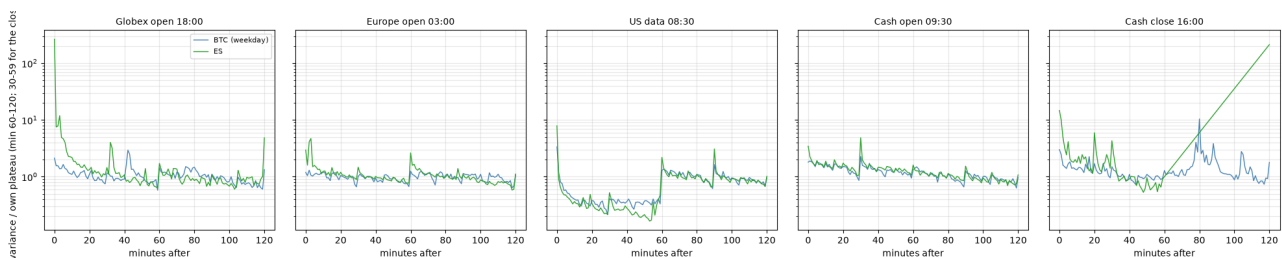


Figure 5 – Mean per-minute variance in the 120 minutes after each event as a multiple of the series' own plateau (minutes 60–120; 30–59 for the close), BTC weekdays (blue) and ES (green), log scale. In the last panel the ES line from minute 60 to 120 is a straight segment across the 17:00–18:00 break, where ES has no data, and carries no information.

event	series	k = 0	k = 1	k = 5	k = 15	k = 30
Globex open 18:00	BTC (weekday)	2.13	1.56	1.65	1.19	1.16
Globex open 18:00	ES	264.35	7.52	4.67	1.48	1.45
Europe open 03:00	BTC (weekday)	1.19	1.05	1.11	1.15	1.01
Europe open 03:00	ES	2.96	1.60	1.58	1.45	1.47
US data 08:30	BTC (weekday)	3.37	1.20	0.59	0.39	0.41
US data 08:30	ES	7.89	0.90	0.48	0.43	0.52
Cash open 09:30	BTC (weekday)	1.81	1.89	1.82	1.68	2.27
Cash open 09:30	ES	3.46	2.25	1.92	1.99	4.87
Cash close 16:00	BTC (weekday)	3.00	2.46	1.69	1.67	1.50
Cash close 16:00	ES	14.70	10.05	4.15	2.24	4.31

Table 7 – Mean per-minute variance k minutes after each event, divided by the same series' own plateau (mean of minutes 60–120 after the event; minutes 30–59 for the 16:00 close because ES has no data in the 17:00–18:00 break). BTC uses weekdays only.

Interpretation. BTC responds to the same events as ES but with far smaller amplitude relative to its own level, because its baseline is already high. The Globex reopen is the extreme case: ES's first minute is $264\times$ its plateau (the whole break's information arrives at once) against $2.1\times$ for BTC, which never stopped trading. The 08:30 release is a one-minute event in both — BTC at $3.4\times$ then $1.2\times$ a minute later; the ratios below 1 afterwards arise because the “plateau” 60–120 minutes after 08:30 is the cash-open hour, which is busier than the minutes following the release. The cash open is different in kind: it is not a spike but a raised level, $\approx 1.8\times$ for BTC and $\approx 2\times$ for ES that persists through the hour, with the 10:00 release adding a further spike ($k = 30$ in the 09:30 row: ES $4.9\times$, BTC $2.3\times$). The 16:00 close lifts BTC to $3\times$ for a minute and $\approx 1.7\times$ for the following half hour. Overall, BTC's US-hours behaviour is a muted version of ES's: the same calendar, roughly a third to a half of the relative amplitude, and a faster return to level.

5. BTC vol response to ES moves

Restricting to minutes in which both series are quoted (873,071 of them), this section conditions BTC's vol over the *next* five minutes on what ES just did: the size of its last one-minute move, in units of the typical ES per-minute vol at that time of day (z_{ES}), and the decile of its deseasonalised variance over the last five minutes.

$ z_{ES} $ bucket (last-minute ES move, in sd units)	n	BTC variance ratio, next 5 min	BTC variance ratio, same minute	BTC vol ratio, next 5 min
[0.0, 0.5)	471,640	0.94	0.93	0.97
[0.5, 1.0)	222,153	1.04	1.04	1.02
[1.0, 2.0)	132,857	1.32	1.33	1.15
[2.0, 3.0)	29,482	1.96	2.01	1.40
[3.0, 5.0)	12,638	2.88	3.00	1.70
[5.0, inf)	4,301	5.48	6.25	2.34

Table 8 – BTC's deseasonalised variance over the next five minutes (mean of x over $t+1\dots t+5$) and in the same minute, bucketed by the size of ES's last one-minute log return measured in units of that minute-of-day's typical ES per-minute vol. 873,071 joint minutes with ES quoted.

decile of ES variance over the last 5 min	n	BTC variance ratio, next 5 min	BTC variance ratio, same minute	BTC vol ratio, next 5 min
D1	87,308	0.76	0.75	0.87
D2	87,307	0.82	0.80	0.91
D3	87,307	0.82	0.80	0.91
D4	87,307	0.84	0.83	0.91
D5	87,307	0.82	0.82	0.91
D6	87,307	0.87	0.84	0.93
D7	87,307	0.95	0.95	0.97
D8	87,307	1.07	1.07	1.03
D9	87,307	1.37	1.35	1.17
D10	87,307	2.77	2.89	1.67

Table 9 – The same BTC quantities bucketed by the decile of ES's deseasonalised variance over the last five minutes.

Interpretation. Both conditionings produce large, monotone effects in the raw data. After an ES minute of more than five typical standard deviations (0.5% of minutes) BTC's variance over the next five minutes averages $5.5\times$ its baseline, i.e. vol $2.3\times$ normal; after a 2–3 sd ES minute, $1.4\times$ normal vol. By ES's trailing 5-minute variance the picture is flat across the bottom six deciles (BTC at 0.76–0.87 of baseline variance) and rises steeply only in the top two (1.37 and 2.77). The tell-tale column is “same minute”: BTC's variance in the minute of the ES move is already as high as it is over the next five (0.93 vs 0.94, 2.01 vs 1.96, 6.25 vs 5.48). BTC does not react to the ES move after the fact; it is moving at the same time, and stays elevated at roughly that level for the next few minutes. This is what one would expect if both are responding to the same news and if BTC's vol is persistent — which makes the next question decisive.

regressors (all logged)	R ²	coefficients
BTC own variance, last 5 min	0.620	0.793
ES variance, last 5 min	0.096	0.359
BTC last 5 min + ES last 5 min	0.623	0.775 (BTC), 0.063 (ES)
BTC last 5 min + ES last 5 min + $ z_{ES} $ (not logged)	0.623	0.775 (BTC), 0.059 (ES), 0.009 ($ z_{ES} $)

Table 10 – OLS of log BTC deseasonalised variance over the next five minutes on trailing five-minute means of the two deseasonalised variances (both in logs) and the size of ES's last-minute move. Minute-by-minute observations, so consecutive rows overlap; the R² values are comparable with one another but the implied standard errors would not be.

Interpretation. Does ES add anything once BTC's own recent vol is known? Essentially not. BTC's own last five minutes explain 62.0% of the variance of its next-five-minute log variance; ES's last five minutes alone explain 9.6%; adding ES to BTC raises R² to 62.3%, with a coefficient of 0.06 on ES against 0.78 on BTC, and adding the size of ES's last-minute move raises it by nothing further (coefficient 0.009). The large conditional effects in Tables 8 and 9 are therefore almost entirely a reflection of BTC's own vol state, which is observed directly and more precisely. For forecasting BTC vol at this horizon, ES is redundant.

5.1 Two-way event study

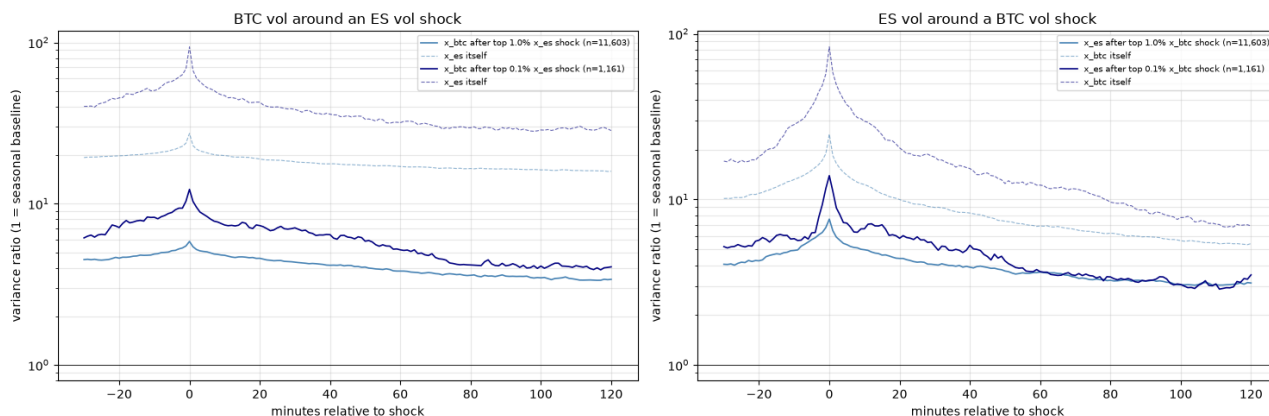


Figure 6 – Left: BTC's deseasonalised variance ratio around the top-1% (light) and top-0.1% (dark) ES variance minutes (solid), with ES's own ratio dashed. Right: the reverse — ES around BTC's variance shocks. Log scale; 1 = seasonal baseline.

shock in	response of	shock set	n	k = -1	k = 0	k = +1	k = +5	k = +15	k = +60
ES	BTC	top 1.0%	11,600	5.47	5.83	5.41	5.03	4.68	3.82
ES	BTC	top 0.1%	1,160	10.4	12.3	10.3	8.32	7.66	5.17
BTC	ES	top 1.0%	11,600	6.75	7.6	6.47	5.33	4.61	3.64
BTC	ES	top 0.1%	1,160	11.6	13.9	11.2	7.19	7.02	3.66

Table 11 – Two-way event study on minutes when both series are quoted: mean deseasonalised variance ratio of the responding series k minutes from a top-1% or top-0.1% variance minute of the other. The notebook's “first minute with excess below 0.25” column is omitted because the response never fell that far within 120 minutes.

Interpretation. When ES has a top-1% vol minute ($\approx 25\times$ its baseline) BTC's variance is $5.8\times$ baseline in the same minute; for a top-0.1% ES minute ($\approx 90\times$), $12\times$. The reverse direction is at least as strong: a BTC shock finds ES at $7.6\times$ (top 1%) or $14\times$ (top 0.1%). Three features stand out. First, both series are already $4\text{--}6\times$ baseline half an hour *before* the other's shock — most of the cross-response is a shared high-vol regime, not the shock itself. Second, the shock-specific bump is modest and contemporaneous: BTC goes from $5.5\times$ at $k = -1$ to $5.8\times$ at $k = 0$ and back to $5.4\times$ at $k = +1$ ($10.4 \rightarrow 12.3 \rightarrow 10.3$ for the extreme set), peaking in the same minute as ES, not the minute after. Third, the decay of the cross-response is slow — still $\approx 3.7\text{--}5\times$ an hour later, never within 25% of baseline inside two hours — because it is the decay of the regime, not of a transient. The symmetry of the two panels argues against a causal story in either direction at the minute scale; common shocks with different amplitudes fit the evidence better.

6. Co-movement of ES and BTC vol, made stationary

Levels of the two vol series share slow regimes, so their correlation mostly says that high-vol weeks are high-vol weeks for both. The question that matters — when ES vol rises unexpectedly, does BTC vol rise with it, and does either lead? — is asked on three stationary transforms of the 5-minute log variance: first differences, AR(12) residuals and HAR residuals (section 1.4), on 219,145 consecutive windows with ES quoted.

transform	corr, same window	β (BTC on ES)	β ES above	β ES below	corr ES(t-1), BTC(t)	corr BTC(t-1), ES(t)	n
levels (log x)	0.303	0.356	0.616	0.152	0.277	0.278	219,145
$\Delta \log x$	0.147	0.238	0.302	0.233	-0.041	-0.038	219,144
AR(12) residuals	0.170	0.267	0.361	0.214	0.034	0.023	219,133
HAR residuals	0.171	0.267	0.367	0.211	0.034	0.021	218,857

Table 12 – Co-movement of ES and BTC deseasonalised 5-minute log variance on 219,145 consecutive windows with ES quoted, for the raw levels and three stationary transforms. “ES above/below” splits on the sign of the ES quantity (above or below its seasonal norm for levels; a rise or fall / positive or negative surprise for the transforms). Two standard errors for a correlation on this sample ≈ 0.004 .

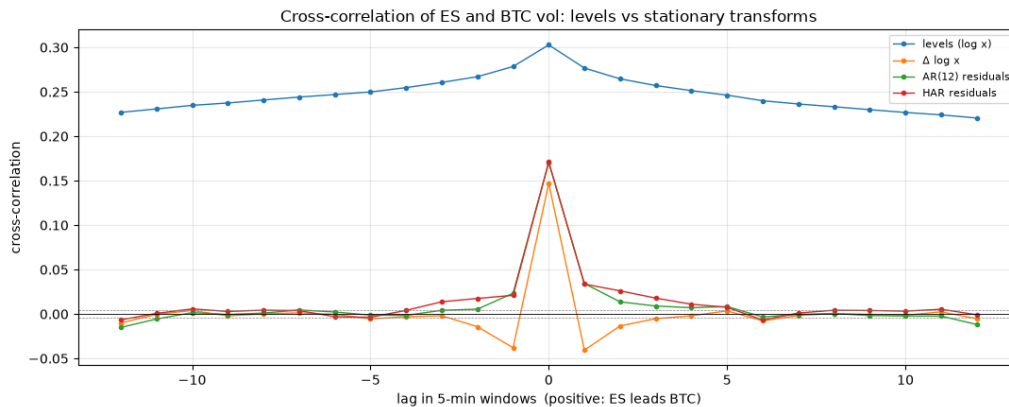


Figure 7 – Cross-correlation of standardised ES and BTC 5-minute log variance at lags of -12 to $+12$ windows (positive lag: ES leads BTC), for levels and the three stationary transforms. Grey dashed lines are $\pm 2/\sqrt{n}$.

Interpretation. Levels correlate at 0.30, and the level CCF (blue) is a broad hump that is still above 0.22 twelve windows out in either direction — the signature of shared persistence. The innovations tell a different story: AR(12) and HAR residuals correlate at 0.17 in the same window, and the CCF collapses to ≈ 0.03 at one window's lead in either direction (ES leading: 0.034; BTC leading: 0.021–0.023) and to noise beyond two. So roughly half of the level correlation is genuine same-window co-movement of surprises, the rest is regime; and the transmission, such as it is, is complete within the window in which it starts. The lead of ES over BTC is measurable (0.034 vs 0.023, on $n \approx 219,000$ with 2 s.e. ≈ 0.004) but at this size it cannot support a rule: knowing that ES vol surprised upward last window moves the expectation of BTC's vol surprise this window by three hundredths of a standard deviation. The first-difference transform shows the expected negative lobes at ± 1 (differencing a noisy series induces them) and the same conclusion.

The response is asymmetric. On HAR residuals, β of the BTC surprise on the ES surprise is 0.37 when ES vol surprises upward and 0.21 when it surprises downward — BTC vol moves about $1.7\times$ as much with an ES vol rise as with an ES vol fall. Figure 9 shows the same thing without a parametric split: the binned relationship is monotone and close to linear, with a steeper right-hand end (from an ES surprise of $+0.6$ to $+1.2$ in log-variance units, BTC's mean surprise rises from 0.13 to 0.37) than left-hand end.

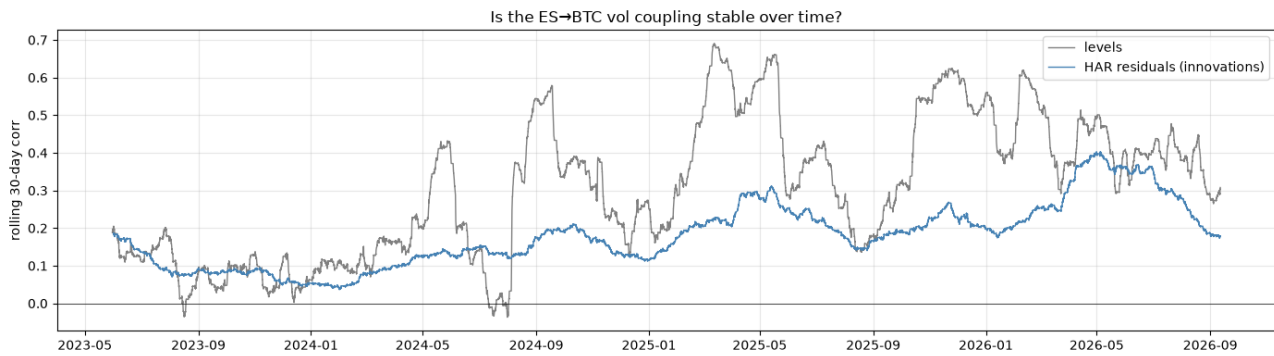


Figure 8 – Rolling 30-day correlation of ES and BTC 5-minute log variance, levels (grey) and HAR-residual innovations (blue), 2023–2026.

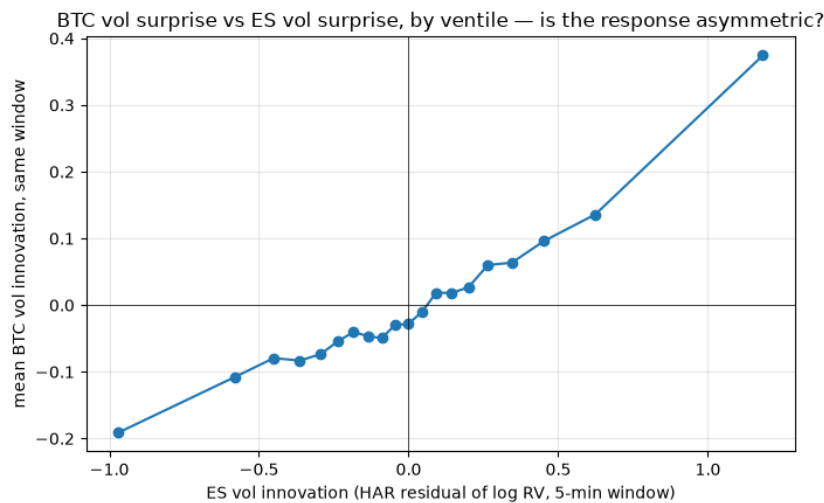


Figure 9 – Mean BTC vol innovation against mean ES vol innovation in each ventile of the ES innovation (HAR residuals of 5-minute log variance).

Interpretation. The coupling has strengthened over the sample. The innovation correlation ran at 0.05–0.10 in 2023 and early 2024, climbed through 2024 to ≈ 0.2 , sat at 0.2–0.3 through 2025, peaked near 0.4 in spring 2026 and stood at ≈ 0.18 in September 2026. The level correlation is far more volatile (from ≈ 0 in mid-2024 to ≈ 0.7 in early 2025), which is what one expects of a statistic dominated by whether the two regimes happened to align in a given month. BTC has become more of a macro asset in its vol dynamics over these three years, but the parameter is not stable enough to be used as a constant; any live use should be on a rolling estimate.

7. Beta from ES to BTC across time frames

This section turns from vol to returns: how much BTC moves when ES moves, over horizons from 10 seconds to a day, and — the question the trade actually needs — whether any of that relationship is available *before* BTC has moved.

horizon	n	β (BTC on ES)	corr	R ²	sd BTC	sd ES	vol ratio BTC/ES	β (ES on BTC)
10s	5,281,026	0.802	0.309	0.096	0.00030	0.00011	2.59	0.119
30s	1,485,493	0.904	0.348	0.121	0.00055	0.00021	2.60	0.134
1min	636,010	0.946	0.371	0.138	0.00082	0.00032	2.55	0.145
5min	125,601	1.019	0.404	0.163	0.00183	0.00073	2.52	0.160
15min	39,893	1.062	0.430	0.185	0.00318	0.00129	2.47	0.174
1h	9,606	1.156	0.462	0.214	0.00639	0.00255	2.50	0.185
4h	2,198	1.163	0.466	0.217	0.01269	0.00508	2.50	0.186
1D	85	1.090	0.592	0.350	0.03202	0.01737	1.84	0.321

Table 13 - Log returns over each horizon on the epoch-aligned 10-second grid, ES quoted throughout and no contract roll inside the interval. $\beta = \text{cov}(r_{\text{BTC}}, r_{\text{ES}}) / \text{var}(r_{\text{ES}})$; sd columns are per-horizon standard deviations.

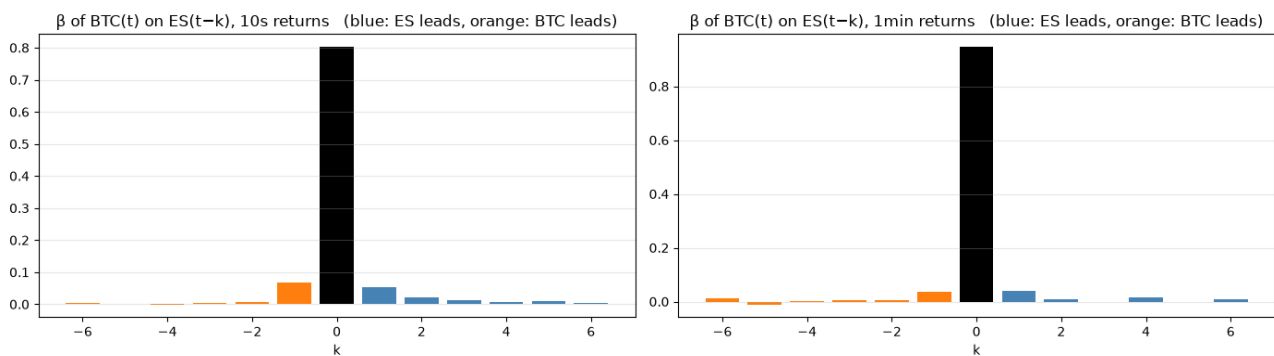


Figure 10 - β of BTC(t) on ES(t-k) for $k = -6 \dots +6$, at 10-second (left) and 1-minute (right) resolution. Black: contemporaneous. Blue ($k > 0$): ES leads BTC. Orange ($k < 0$): BTC leads ES.

Interpretation. BTC's return β on ES is about 1 at the 5-minute horizon (1.02), with correlation 0.40: a 1 bp ES move is matched on average by a 1 bp BTC move, and ES accounts for 16% of the variance of BTC's 5-minute return. The vol ratio BTC/ES is a steady 2.5 across horizons, so the correlation of 0.40 and β of 1 are two views of the same fact. β rises from 0.80 at 10 seconds through 0.95 at 1 minute to 1.02 at 5 minutes and 1.16 at 1 hour; correlation from 0.31 to 0.46. About 80% of the 5-minute co-movement is therefore already present at 10 seconds, and the remainder builds over the next minute or two. Part of the low 10-second correlation is measurement: microstructure noise in 1-second closes and a 0.25-point ES tick dilute correlation at the shortest horizons. The 1-day row ($n = 85$) has a much higher correlation (0.59) but too few observations to weight heavily.

The lead-lag bars settle the direction question at this resolution: at 10 seconds the contemporaneous β is 0.80, the $k = +1$ bar (ES leading by 10 s) is ≈ 0.055 , the $k = -1$ bar (BTC leading by 10 s) ≈ 0.07 , and everything else is below 0.02; at 1 minute the $k = \pm 1$ bars are ≈ 0.04 each. The lagged terms are small, roughly symmetric and sum to about the difference between the 10-second and 5-minute β . **There is no measurable lead of ES over BTC at 10-second resolution:** either both react to common information inside the same 10 seconds, or the lead is shorter than the grid. Either way it is not something a 5-minute market participant can act on.

session	n	β	corr	BTC vol (ann.)	ES vol (ann., 24/7 clock)
Globex reopen 18-19	890	0.330	0.240	0.712	0.518
Asia 19-03	17,433	1.217	0.357	0.564	0.166
Europe 03-08:30	28,972	0.868	0.348	0.455	0.182
US pre 08:30-09:30	8,868	1.100	0.430	0.569	0.223
Cash open 09:30-10:30	10,014	1.118	0.460	0.864	0.355
US midday 10:30-14	34,565	1.134	0.439	0.640	0.248
US pm 14-16	19,485	0.954	0.414	0.552	0.239
Post-close 16-17	5,174	0.661	0.286	0.581	0.252

Table 14 - The 5-minute β and correlation by session (ET). Both vols are annualised on the 24/7 clock so they are comparable across rows; the ES figures are therefore lower than the trading-time convention would give.

ES move bucket (sd)	n	β	corr	P(same sign)	mean $r_{\text{BTC}} \mid \text{ES up}$	mean $r_{\text{BTC}} \mid \text{ES down}$
[0.0, 0.5)	74,479	1.131	0.142	0.566	+0.00019	-0.00018
[0.5, 1.0)	30,236	1.097	0.325	0.670	+0.00056	-0.00057
[1.0, 2.0)	15,698	1.128	0.503	0.770	+0.00106	-0.00115
[2.0, 3.0)	3,328	1.116	0.643	0.854	+0.00191	-0.00197
[3.0, inf)	1,860	0.892	0.749	0.920	+0.00349	-0.00320

Table 15 - 5-minute windows bucketed by the size of the ES move in units of its own standard deviation (1 sd $\approx$ 7.3 bp). "P(same sign)" is the fraction of windows in which BTC's return over the same window had the sign of ES's.

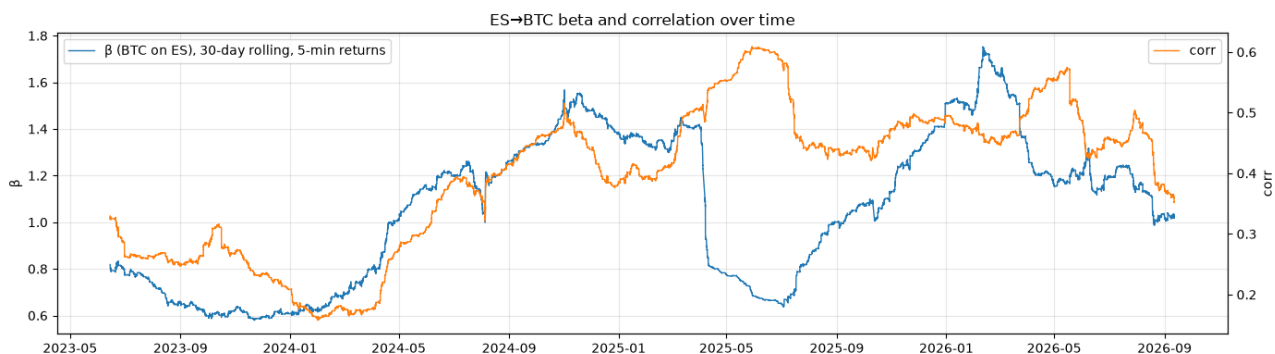


Figure 11 - Rolling 30-day β of BTC on ES (blue, left axis) and correlation (orange, right axis), 5-minute returns, ES-open windows.

Interpretation. By session, correlation is highest in US hours (0.43–0.46 from 08:30 to 14:00) and lowest in the Globex reopen hour (0.24) and after the close (0.29). β follows a different pattern because it also depends on the vol ratio: it is highest in the Asian session (1.22), where ES vol is very low (17% annualised on the 24/7 clock) relative to BTC's (56%), and lowest at the reopen (0.33, on only 890 windows, where ES's reopen jump has no BTC counterpart). By ES move size, β is remarkably stable — 1.10–1.13 for every bucket up to 3 sd, 0.89 beyond — while correlation rises from 0.14 for sub-half-sd moves to 0.75 for moves over 3 sd, and P(same sign) from 0.57 to 0.92. The conditional means are symmetric in sign (e.g. +10.6 bp / –11.5 bp for a 1–2 sd ES move; +34.9 / –32.0 bp beyond 3 sd), so there is no directional asymmetry to exploit. When ES moves more than 3 sd ($\approx$ 22 bp) in a five-minute window, BTC ends the same window with the same sign 92% of the time — but this is a same-window statement, which only becomes known as the window closes. Figure 11 shows that none of these numbers is a constant: the 30-day β ranged from 0.58 (late 2023) to 1.75 (early 2026), and in April 2025 it fell to $\approx$ 0.7 while correlation rose to 0.6 — ES vol spiked more

than BTC's during the tariff episode, so BTC moved less per point of ES even though the two were more tightly linked.

7.1 Within a 5-minute window: does ES's move so far predict BTC's remaining move?

This is the number that matters for an up/down market entered mid-window. After k minutes of a window, BTC's log return over the remaining $5 - k$ minutes is regressed on ES's return over the first k minutes and, separately and jointly, on BTC's own return over the first k .

k (minutes elapsed)	β : BTC rest on ES first k	t- stat	R ² ES only	β : BTC rest on BTC first k	R ² BTC only	R ² both	P(BTC rest same sign as ES so far)	... when ES so far > 2 sd	n (> 2 sd)	sd BTC rest	sd BTC first k
1	0.0585	5.02	0.0002	-0.0235	0.0002	0.0006	0.5012	0.4946	5,000	0.0016	0.0009
2	0.0381	5.02	0.0002	-0.0055	0.0000	0.0003	0.5005	0.5017	5,529	0.0013	0.0012
3	0.0148	2.91	0.0001	-0.0063	0.0001	0.0002	0.4984	0.5018	5,518	0.0011	0.0014
4	0.0087	2.77	0.0001	-0.0017	0.0000	0.0001	0.4947	0.5036	5,699	0.0007	0.0016

Table 16 – Within a 5-minute window, ES quoted (137,493 complete windows). After k minutes, BTC's return over the remaining $5-k$ minutes is regressed on ES's and on BTC's own return over the first k minutes. Returns are log returns; 0.0016 = 16 bp.

Interpretation. The answer is a clear negative. After one minute, β of BTC's remaining four-minute return on ES's first-minute return is 0.059: a one-standard-deviation ES move in the first minute (≈ 3 bp) predicts 0.2 bp of BTC over the remaining four minutes, against a remaining-return standard deviation of 16 bp. The t-statistic of 5 on 137,000 non-overlapping windows confirms the coefficient is not zero; the R² of 0.0002 confirms it is useless. The coefficient shrinks further as the window progresses (0.038, 0.015, 0.009). The probability that BTC's remaining move has the sign of ES's move so far is 0.501, 0.501, 0.498, 0.495 — and conditioning on a large ES move so far (more than 2 sd, $\approx 5,000$ –5,700 windows) gives 0.495, 0.502, 0.502, 0.504, indistinguishable from a coin flip. This is exactly what section 7's contemporaneous $\beta \approx 1$ and flat lead-lag bars imply: whatever ES has done in the first k minutes, BTC has already matched, and the remaining move is fresh. BTC's own return so far is equally uninformative about its remaining return ($\beta \approx -0.02$, a faint reversal; $R^2 \leq 0.0002$).

What is informative at a mid-window entry is not in this table but follows from it: the current price relative to the window open, and the standard deviation of the remaining move (16, 13, 11 and 7 bp after 1, 2, 3 and 4 minutes on average, and scalable by the vol multiple of section 3). With no drift and no momentum, the fair probability that a window currently up by d finishes up is approximately $\Phi(d/\sigma_{\text{rest}})$: a window 10 bp up after three minutes, with $\sigma_{\text{rest}} \approx 11$ bp, is $\approx 82\%$ to close up.

8. Implications for the 5-minute up/down trade

- **Direction cannot be forecast from price history, and not from ES either.** P(up) is 0.498 unconditionally; last-window sign, BTC's own move so far in the window and ES's move so far in the window all leave the sign of the remaining move at $50 \pm 0.5\%$. Any edge in this market has to come from the price of the contract relative to the current position of BTC within the window, from vol, or from settlement mechanics — not from a directional signal.

- **Vol is the forecastable input, and it varies by a factor of four to six.** Time of day and weekday/weekend set the baseline (≈ 8 bp per window on a weekend night to ≈ 31 bp in the cash-open window); the current vol state multiplies it by 0.3–2.2 (Table 5); a 30-minute trailing mean captures most of the forecastable part (R^2 0.56 versus 0.59 for the full HAR). A fair mid-window price $\Phi(d/\sigma_{\text{rest}})$ is only as good as σ_{rest} , and this report says σ_{rest} is well estimated from BTC alone.
- **ES is not needed for the vol estimate.** It contributes 0.3 percentage points of R^2 beyond BTC's own last five minutes and its vol surprises lead BTC's by a correlation of 0.03. Where ES might help is as a proxy for scheduled-event risk (08:30, 09:30–10:30, 10:00, 16:00), but the calendar itself does that job directly.
- **Respect the noise floor.** One window in ten ends within 1 bp of its start and one in five within 2 bp. In those windows the resolution depends on print timing and on the Chainlink/Binance discrepancy, not on anything modelled here; a fair-value model should treat a small $|d|$ late in the window as near-50/50 rather than trusting $\Phi(d/\sigma_{\text{rest}})$ with a tiny σ_{rest} .
- **After a violent minute, stay in the high-vol regime for the hour.** Excess vol decays as a power law: $3\times$ baseline vol 30 minutes after a top-1% minute, $2.3\times$ two hours later. This holds identically whether or not ES is open.
- **The weekday/weekend and US-session distinctions are large enough to require separate parameters.** Weekday variance is $2\text{--}3\times$ weekend overnight and $7\text{--}25\times$ at US event times; a single seasonal profile would misprice both.

9. Caveats

- **Binance spot is not the settlement price.** The market resolves on Chainlink BTC/USD, an aggregate of several exchanges updated on its own schedule. Levels and 5-minute returns are very close, but the boundary values that decide a window can differ by more than the 1–2 bp that decides 10–20% of windows. The base rates in section 2 ($P(\text{up})$, $P(|r_5| < 1 \text{ bp})$, ties) should be re-estimated on the Chainlink series before they are used for pricing.
- **1-second realised variance contains microstructure noise.** Bid–ask bounce in 1-second closes (BTC) and the 0.25-point tick (ES, partly mitigated by the microprice) add a positive bias to RV that depends on activity, and dilute correlations at the 10-second horizon. The deseasonalised ratios $\times$ are less affected than the levels because the bias is largely absorbed into the seasonal mean.
- **Overlapping samples.** The minute-by-minute regressions of section 5 use rolling five-minute means, so consecutive observations overlap heavily; their R^2 values are valid for comparison but no standard errors are quoted for them. The HAR and within-window regressions use non-overlapping windows. The event studies average over shocks that can be minutes apart and therefore share post-shock minutes.
- **In-sample fits.** The ACF shape fits, the decile table, the shock-decay fits and all the β and correlation tables are in-sample descriptions. Only the HAR forecasts (Table 4) were checked out of sample. The exponential half-lives of Table 6 are poor fits and included for completeness only.
- **Deseasonalisation uses means.** The baseline is the mean variance at each minute of day, which is dominated by the largest days; typical windows sit at 0.6–0.7 of it. Median-based multiples are reported alongside the mean-based ones for this reason.

- **Weekday/weekend classification by ET calendar day** mixes Friday evening (ES closed) into “weekday” and Sunday evening (ES open) into “weekend” between 18:00 and midnight; the ratio in Figure 4 is only clean from 00:00 ET onward.
- **A single, eventful three-year regime.** The sample spans the spot-ETF approval (January 2024), the November 2024 election and the April 2025 tariff shock; the rolling plots show the ES-BTC coupling roughly doubling over it. Means over ≈ 870 weekdays can be moved by a few days, which is the likely origin of the unattributed spikes at 07:00 and 17:15–17:30 on weekdays and near 16:00 and 21:00 on weekends.
- **Thin cells.** The 1-day β row has 85 observations, the Globex-reopen session 890 windows, and the >3 sd ES bucket 1,860 windows; these are indicative only.